

Thesis Mid Report for the Degree of Master of Science in Computer Science

Reducing False Alarms in IoT Networks: A Deep Learning Based Intrusion Detection System



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List of Abbreviation/Acronyms

ADYSN	Adaptive Synthetic Sampling
Bi-LSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Networks
FNR	False Negative Rate
FPR	False Positive Rate
GRU	Gated Recurrent Unit
MITM	Man-in-the-Middle
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique

CHAPTER 1

INTRODUCTION

The Internet of Things (IoT) has changed modern life by connecting billions of devices and are now present in every aspect of our daily lives. Examples range from our homes, with smart appliances and entertainment systems, to smart cities and their hidden infrastructure, like pedestrian and road sensors. The smart grid, self-driving car systems, smart medical devices, industrial control systems, and robotics are just some areas where IoT devices are used every day. The data collected by these devices needs storage and analysis. To gain insights from this data and support smart applications, methods like machine learning (ML) are commonly used. Detecting cyber attacks is part of these smart applications. With the ongoing rise in cyber attacks targeting IoT systems, monitoring this large amount of data for attack detection is essential. However, this can only be done through automated methods that rely on ML and deep learning (DL). DL is a subset of ML that has gained popularity in various fields, including science, finance, medicine, and engineering. Using DL for intrusion detection has also become more common because it allows for deeper analysis of network traffic and more accurate identification of anomalies compared to traditional ML methods. This unprecedented scale and heterogeneity, however, have simultaneously created an enormous and vulnerable attack surface. These low-power systems are constantly exposed to a spectrum of sophisticated cyber threats, such as Distributed Denial-of-Service (DDoS), Man-in-the-Middle (MITM), data injection, and ransomware attacks.

In these networks, intrusion detection systems (IDS) are a crucial defense mechanism because they can identify unusual activity that traditional security measures would miss. These security mechanisms monitor network traffic and device behavior to identify malicious activity that traditional perimeter defenses, like firewalls, often fail to detect. Recently, deep learning techniques have emerged as the most promising solutions, reporting superior overall detection accuracy (often above 98%) compared to traditional machine learning classifiers. Nevertheless, despite the tremendous advancements, the high false positive rate (FPR) remains one of the most significant present issues with IoT IDS.

Even though state-of-the-art models like CNN-GRU [1] and Random Forest [2] have reported high overall detection accuracy of more than 98%, their FPRs, ranging between 0.50% and 0.75%, still equate to a large number of false alarms.

In reality, in large IoT installations, even a low FPR can result in thousands of false alarms per day, causing resource waste, accelerated battery drain, network congestion, and analyst fatigue. Furthermore, frequent false positives undermine operators' trust in automatic intrusion detection systems, leading them to disregard warnings or completely disable detection systems, increasing their vulnerability to successful cyberattacks.

To address these critical challenges, this research introduces a novel optimized Bidirectional Long Short-Term Memory (BiLSTM) architecture specifically designed for

IoT intrusion detection, with particular emphasis on reducing false positive rates while maintaining high detection sensitivity across diverse attack categories.

1.1 Background

The security landscape for the Internet of Things (IoT) presents significant challenges. Traditional Intrusion Detection System (IDS) models are often unsuitable because of the demanding computational and energy constraints inherent to most IoT devices [4]. Furthermore, the heterogeneous nature of IoT traffic—which includes everything from routine sensor telemetry to malicious data like botnet attacks—differs substantially from typical network data [3]. This variability makes it difficult to create security models with good generalization, leading to poor performance when trying to identify diverse threats.

Current research favors Deep Learning and Hierarchical Models for improved IoT security. Deep learning often utilizes hybrid models like CNN-GRU [1], RNN-GRU [12], and LSTM-GRU [10], which are highly effective at extracting complex features from diverse IoT data. However, a limitation of these models is their reduced capability in detecting bidirectional temporal patterns within the data. Alternatively, Hierarchical Models have shown promise, with one study demonstrating that Multi-Layer Perceptrons (MLPs) significantly reduced false alarms by 87.52%. Despite this success, the general applicability of hierarchical models is limited by a lack of IoT-specific validation in their deployment and testing environments [7].

To address these shortcomings, this study leverages a Bidirectional Long Short-Term Memory (BiLSTM) network as the core of the proposed intrusion detection framework. The BiLSTM architecture processes input sequences in both forward and backward directions, enabling it to learn comprehensive temporal dependencies and recognize complex attack behaviors that may otherwise go undetected. By capturing context from both directions, the model achieves a deeper understanding of IoT traffic patterns, improving its ability to detect diverse attack types.

Furthermore, the BiLSTM model is optimized using Bayesian optimization to systematically fine-tune its hyperparameters for maximum detection performance and minimum false alarm rate. This optimization ensures that the model operates efficiently even under the computational constraints typical of IoT devices.

Overall, the proposed approach enhances IoT security by utilizing BiLSTM’s bidirectional temporal learning and Bayesian-driven optimization to deliver accurate, generalized, and resource-efficient intrusion detection, effectively strengthening the reliability of IoT networks.

1.2 Statement of the problem

The high false positive rate (FPR) remains a critical challenge for Internet of Things (IoT) intrusion detection systems (IDS), significantly diminishing their real-world efficacy despite advancements in their underlying models. Even systems boasting high detection accuracy, often over 98%, frequently struggle to adapt to the dynamic, heterogeneous, and ever-changing nature of IoT network traffic. This varied traffic, originating from a multitude of devices, protocols, and behavioral patterns—many of which are benign but unique—can easily be misclassified as malicious anomalies by the anomaly-based detection techniques commonly employed in IoT IDS. This results in a continuous flood of false alarms, which leads to alert fatigue among security analysts, making it difficult to differentiate genuine threats from harmless events, and consequently increasing the risk of overlooking a true attack. [1][2]

1.3 Research objectives

To perform a thorough evaluation of both baseline and enhanced BiLSTM models to measure improvements in intrusion detection effectiveness across various aspects.

To Develop an optimized intrusion detection system that achieves class-specific performance targets, with particular focus on reducing false alarms while maintaining high detection accuracy for all attack types.

1.4 Significance/Rationale of the study

This study presents a Bi-LSTM framework to address the important problem of high false positive rates (FPR) in IoT intrusion detection systems (IDS). Because they rely on non-IoT datasets and unidirectional learning, traditional IDS models like CNN-GRU and Random Forest overproduce false alarms that waste resources, overwhelm analysts, and delay adoption. In addition to efficiency, sustainability, security, and reliability all depend on lowering FPR in IoT IDS systems.

CHAPTER 2

LITERATURE REVIEW

2.1 The Foundation: IoT Datasets and Traditional ML Models

The development of realistic IoT-specific datasets has been a crucial step forward for IDS research. The ToN-IoT dataset [9, 14], introduced by Moustafa et al., is a comprehensive benchmark that merges data from network traffic, device telemetry, and operating system logs, effectively capturing the heterogeneous nature of real IoT systems. This dataset has become a cornerstone for evaluating IoT IDS. Early studies leveraging such datasets often employed traditional machine learning models. For instance, Gad et al. [11] used XGBoost on the ToN-IoT dataset, achieving high accuracy by applying SMOTE for class imbalance and Chi-squared for feature selection. Similarly, Rana and Singh [2] achieved a 99.4% accuracy with Random Forest on the NSL-KDD dataset. However, a significant limitation of these models is their inability to capture complex spatial-temporal patterns in IoT traffic. Furthermore, as noted by Gad et al. [11], such approaches lack integrated explainability and their performance on non-IoT datasets [2] raises concerns about their generalizability to IoT-specific attack vectors.

2.2 Deep Learning and Hybrid Models for Enhanced Detection

Using feature importance analysis in binary and multi-class classification frameworks, Rana and Singh proposed a Random Forest-based IDS with a 99.4% accuracy and decreased the false alarm rate to 0.50% on the NSL-KDD dataset. Even though it works well for conventional networks, using a non-IoT dataset makes the model unscalable because it doesn't have any attack patterns specific to IoT and has a high failure rate when used in IoT environments [2].

Mijalkovic and Spognardi concentrated on the application of deep learning and artificial neural networks (ANN) and deep neural networks (DNN) for minimizing the false negative rates (FNR) of intrusion detection systems (IDS). They addressed the problems of minority-class attack detection, especially uncommon Internet of Things-specific attacks. In their approach, they experimented with imbalanced learning techniques on datasets such as UNSW-NB15 (non-IoT) and NSL-KDD and utilized DNN and ANN for multi-class classification. Similar to IoT-specific attacks, the minority class recall of the experiments was low and their generalization to IoT traffic was poor, despite having an improvement in their FNR. Use of Non-IoT dataset and not prioritizing the minimization of FPR were two of the significant drawbacks [3].

By combining temporal sequence modeling and spatial feature extraction, Henry et al. developed a CNN-GRU-based intrusion detection system (IDS) that demonstrated 98.73%

accuracy and a low FPR of 0.0075 on the CICIDS-2017 dataset. Aside from its limited generalizability to IoT networks due to its evaluation on a non-IoT dataset, the model's two main shortcomings are its inability to capture long-term dependencies and its imbalanced attack classification with few attack classes. [1].

Siganos et al. enhanced analyst confidence in model outputs by integrating SHAP-based Explainable AI (XAI) with an Artificial Intelligence (AI)-driven Intrusion Detection System (IDS) for the Internet of Things. They acknowledged current challenges such as false alarms and greater explainability. They evaluated on IEC 60870-5-104 protocol data (industrial IoT) and the solution emphasized using SHAP for visualization of feature importance. Although decision-making was facilitated by Explainable Artificial Intelligence (XAI), no impact was observed on false positive rates (FPR), and protocol-specific anomalies led to periodic false alarms. The study was limited by the exclusive consideration of a single industrial protocol and the absence of mechanisms for reducing false alarms. The results highlighted the imperative to incorporate XAI in Internet of Things Intrusion Detection Systems (IoT IDS) for mitigating FPR issues [5].

On the basis of a two-stage MLP model, Moore et al. proposed an IDS that achieved 73.44% recall, 0.59% FPR, and a 87.52% decrease in false alarms on the CIDDS-001 dataset. The high-risk alerts were prioritized, and minority-class detection was improved through multi-task learning. Although the model was successful in decreasing false positives, its application was restricted to IoT environments as it was evaluated solely on non-IoT datasets. Its low performance against the attacks that are uncommon further indicates that it requires additional fine-tuning to deal with Internet of Things challenges [7].

By combining Bi-LSTM to enable bidirectional temporal modeling with SHAP and LIME for interpretability, Sivamohan and Sridhar created a BiLSTM-XAI model that produced a 98.2% accuracy rate and a 0% false positive rate (FPR) on the NSL-KDD and Honeypot datasets. The suggested model improved feature selection by utilizing Krill Herd Optimization (KHO). Despite the impressive results, the model's applicability to the current IoT scenario is questionable because it was tested on outdated, non-IoT datasets rather than being compared to the current, IoT-specific attack patterns. Further testing on actual IoT traffic is necessary to determine its feasibility in real-world scenarios. [8].

Using the ToN-IoT dataset, Gad et al. proposed a distributed IDS using XGBoost for binary and multi-class classification of IoT-based attacks. The study applied the Synthetic Minority Over-sampling Technique (SMOTE) for handling class imbalance and feature selection using the Chi-squared (χ^2) approach to reduce the feature set to 20 features, successfully mitigating serious issues related to data quality. The training efficacy and classification performance were greatly improved by tuning the preprocessing pipeline. When testing generalization to other attack types such as Ransomware and DDoS, the optimized model performed significantly better than other machine learning baselines, with higher accuracy and F1-scores.

However, using a traditional machine learning model like XGBoost compromises the system's capacity to recognize complex spatial-temporal patterns associated with IoT traffic. Furthermore, the study does not incorporate explainability frameworks and deep learning techniques, which are increasingly important in contemporary IDS research. The model would be enhanced by incorporating robust temporal modeling and interpretability capabilities, despite its effectiveness in detecting anomalies in IoT environments [11].

Using the CICIDS-2017 dataset, Al-Kahtani et al. proposed a hybrid LSTM-GRU model in conjunction with a PSO-GA (Particle Swarm Optimization–Genetic Algorithm) feature selection technique, which achieved 98.86% accuracy. Their approach successfully uses deep sequential models to achieve robust attack detection and evolutionary algorithms to choose the best subset of features. The model is computationally costly and has not been tested on IoT-specific traffic, which makes it unsuitable for deployment on resource-constrained IoT devices, even though it achieves state-of-the-art performance in traditional network environments. Consequently, the framework is not directly appropriate for the requirements of IoT security, even though it excels at feature selection and detection [10].

A hybrid RNN-GRU approach to IoT intrusion detection was proposed by Khan et al. and outperformed state-of-the-art methods, achieving 99% accuracy on TON-IoT network flow data. With extensive data engineering to fine-tune features, their method used GRU for long-term dependency capturing and RNN for sequence modeling. With 99% accuracy for the network flow and 98% accuracy for the application layer, the model outperformed cutting-edge methods in identifying frequent IoT attacks. However, two significant limitations of the TON-IoT dataset are its high computational requirements for devices with limited resources and its limited representation of real-world attack vectors. The hybrid model exhibits promise despite these obstacles, but more testing against various attack scenarios is necessary [12].

Using the TON-IoT dataset, Alabbadi & Bajaber combined Explainable AI (XAI) methods such as SHAP and LIME with 1D-CNNs and DNNs to detect IoT intrusions with 99.24% accuracy. Their method showed how XAI can improve IoT attack classification transparency. The strategy used pre-trained TabNet for effective tabular data processing and 1D-CNNs for feature extraction from IoT data streams. For specific attack classes, the results showed 100% accuracy for CNN/DNN, while SHAP/LIME provided useful feature importance insights. However, the study had certain shortcomings, such as a lack of examination of adversarial robustness for XAI outputs and a restricted emphasis on minimizing the false positive rate (FPR). The model is useful for detecting IoT attacks, but it is not FPR- and security-optimized, according to the conclusion [4].

Gardner, Ashadevi, Swersky, Wang, and Poloczec (2019) presented an extension of Bayesian Optimization (BO) to handle inequality constraints in black-box optimization problems, a common requirement in real-world machine learning and engineering applications. Traditional BO assumes unconstrained optimization, which limits its

applicability when feasible solutions must satisfy safety, cost, or performance boundaries. The authors proposed a Gaussian Process (GP)-based model that jointly represents both the objective function and the constraint functions, capturing their uncertainty through probabilistic inference. By defining an expected constrained improvement (ECI) acquisition function, the method efficiently balances exploration (searching for promising areas) and exploitation (refining known feasible regions). [17]

Chen, Yang, and Wang (2023) proposed an Adaptive Bi-LSTM with Focal Loss intrusion detection system (IDS) designed to overcome class imbalance challenges in multi-class IoT attack detection. The model leverages a Bidirectional Long Short-Term Memory (Bi-LSTM) network to capture both forward and backward dependencies in sequential network traffic data. Its core innovation lies in the Adaptive Focal Loss (FL), which dynamically adjusts the loss weights based on class distribution and prediction confidence. This mechanism reduces the influence of easily classified normal samples while emphasizing the learning of minority and hard-to-classify attack types, thereby enhancing model balance and sensitivity. Evaluated on the UNSW-NB15 dataset, the model achieved a False Positive Rate (FPR) of 0.52% and a False Negative Rate (FNR) of 0.88%, demonstrating notable improvement in detecting rare attack classes. Moreover, the adaptive mechanism maintained robust performance across varying imbalance ratios, confirming its generalization capability [18].

Kumar, Singh, and Patel (2022) proposed a Weighted Bi-LSTM model to reduce false negatives in IoT intrusion detection systems. Traditional IDS models often struggle with imbalanced datasets, where rare attack types are misclassified as normal traffic. To address this, the authors introduced a class-weighted loss function that assigns higher penalties to misclassifications of minority attack classes, thereby improving sensitivity to less frequent attacks. The model employs a Bidirectional Long Short-Term Memory (Bi-LSTM) architecture to analyze IoT network traffic sequences, capturing both forward and backward temporal dependencies. Experiments were conducted on the UNSW-NB15 and BoT-IoT datasets, evaluating the model's robustness across different attack distributions. On the UNSW-NB15 dataset, the Weighted Bi-LSTM achieved an Accuracy of 98.7%, a False Positive Rate (FPR) of 0.65%, and a False Negative Rate (FNR) of 0.72%, outperforming baseline models such as standard Bi-LSTM and CNN-LSTM. The approach notably improved detection for minority attacks (e.g., Exploits, DoS, and Reconnaissance) without sacrificing overall classification stability [19].

Rodriguez, Martinez, and Lopez (2023) proposed a Constrained Deep Learning (CDL) framework for IoT intrusion detection systems (IDS) designed to maintain bounded False Positive Rates (FPR) without compromising detection performance. Recognizing that high FPRs are a major obstacle in deploying IDS models in real-world IoT environments, the authors introduced a constraint-based optimization mechanism that explicitly integrates an upper bound on FPR into the model's training objective. The proposed CDL model builds

upon a deep neural network architecture, incorporating both feature selection and regularization constraints to ensure stable learning under noisy and imbalanced traffic conditions. This approach effectively balances the trade-off between detection sensitivity (reducing false negatives) and false alarm control (limiting false positives). Evaluations conducted on the BoT-IoT and CICIDS2017 datasets showed that the CDL-based IDS achieved an Accuracy of 98.4%, a False Positive Rate (FPR) below 1% (bounded at 0.95%), and a False Negative Rate (FNR) of 1.12%. Compared to unconstrained deep models, the CDL approach exhibited more stable and predictable performance, particularly under changing network conditions and varying imbalance levels. [21]

Smith, Brown, and Davis (2023) presented a comprehensive survey and comparative analysis of deep learning (DL)-based intrusion detection systems (IDS) designed for IoT networks. Published in *ACM Computing Surveys*, this study systematically reviewed over 80 state-of-the-art IDS models—including CNNs, RNNs, Bi-LSTMs, GANs, and hybrid deep learning frameworks—to evaluate their effectiveness, scalability, and robustness in handling the complex and imbalanced nature of IoT traffic data. The paper identified that models leveraging temporal learning (e.g., LSTM and Bi-LSTM) and attention mechanisms outperform static architectures in recognizing evolving attack behaviors. Moreover, the survey emphasized the significance of loss function design (e.g., Focal Loss, Weighted Loss) and data preprocessing strategies (normalization, feature selection, and augmentation) in improving model generalization and handling minority attacks. Experimental comparisons on benchmark datasets—UNSW-NB15, BoT-IoT, and ToN-IoT—showed that top-performing DL models achieved Accuracy between 98–99%, with False Positive Rates (FPR) as low as 0.4–0.9% and False Negative Rates (FNR) under 1.2%. The study concluded that while DL-based IDSs provide strong detection capabilities, real-world deployment challenges such as computational cost, explainability, and adaptability to new attacks remain open research issues. [22]

Wang, Li, and Zhao (2022) introduced a Dual-Constraint Bi-LSTM model for intrusion detection in IoT networks, emphasizing low false alarm rates while maintaining high detection accuracy. The proposed approach integrates two key constraints during model training: a regularization constraint to prevent overfitting and a false alarm constraint that penalizes excessive false positives. This dual-objective optimization ensures that the model achieves both robust generalization and operational reliability in real-world IoT environments. The Bidirectional Long Short-Term Memory (Bi-LSTM) architecture captures bidirectional temporal dependencies in IoT traffic sequences, enabling better detection of subtle attack patterns that evolve over time. The model was evaluated on the BoT-IoT and CICIDS2017 datasets, comparing its performance with traditional Bi-LSTM, CNN-LSTM, and GRU-based models. Experimental results demonstrated that the Dual-Constraint Bi-LSTM achieved an Accuracy of 99.2%, a False Positive Rate (FPR) of 0.41%, and a False Negative Rate (FNR) of 0.67%, outperforming existing baselines. Additionally, the model maintained consistent detection accuracy under varying network loads and attack distributions, showcasing its robustness and adaptability. [23]

Banaamah and Ahmad (2022) propose a deep learning-based framework for intrusion detection in Internet of Things (IoT) environments. The study addresses the limitations of traditional machine learning IDS models that rely heavily on handcrafted features and often fail to capture complex, nonlinear attack patterns in IoT traffic. The authors used the BoT-IoT dataset, which contains a mix of normal and malicious traffic patterns representative of real IoT systems. Models were evaluated using standard metrics such as accuracy, precision, recall, and F1-score. The LSTM model achieved the best overall performance with approximately 99.8% accuracy, outperforming CNN and GRU models. LSTM achieved the highest performance with Accuracy = 99.8%, Precision = 0.998, Recall = 0.997, F1 = 0.998, FPR $\approx$ 0.002, and FNR $\approx$ 0.003. This result highlights the advantage of sequence-based models in capturing temporal dependencies in IoT traffic data. [15]

Sadhwani, Khan, Muthalagu, and Pawar (2023) proposed a BiLSTM-CNN Hybrid Intrusion Detection System (IDS) designed to enhance detection accuracy and reduce false alarms in IoT network security. The hybrid model effectively combines the feature extraction capability of Convolutional Neural Networks (CNN) with the temporal sequence modeling power of Bidirectional Long Short-Term Memory (Bi-LSTM) networks. In the proposed architecture, CNN layers first capture spatial correlations and local traffic patterns, while the Bi-LSTM layers model long-term dependencies within network flow sequences, allowing for a more comprehensive understanding of attack behavior. This dual-layer synergy enables the system to detect both instantaneous and evolving intrusion patterns. The model was tested on benchmark IoT datasets such as BoT-IoT and CICIDS2017, achieving an Accuracy of 99.3%, a False Positive Rate (FPR) of 0.38%, and a False Negative Rate (FNR) of 0.59%. Compared to single-architecture models, the BiLSTM-CNN hybrid demonstrated higher precision and recall, particularly for minority attack classes, and maintained computational efficiency suitable for real-time IoT applications. [16]

2.2 Research Gap

- Most IDS are trained on non-IoT datasets (CICIDS-2017, NSL-KDD), existing IDS models [1][2] have poor generalization to IoT traffic patterns (e.g., botnets, DDoS on constrained devices).
- State-of-the-art models [1][2] report FPRs of 0.50 to 0.75%; however, in IoT environments, these rates increase because of noise and heterogeneous traffic.
- Previous research ignores cost-sensitive tuning to minimize FPR under stringent FNR constraints (e.g., $\leq 1\%$) in favor of accuracy or FNR reduction [3].

2.3 Theoretical Background

2.3.1 Internet of Things (IoT) and Network Security

The Internet of Things (IoT) represents a rapidly expanding network of interconnected devices that communicate and share data autonomously. These devices, ranging from sensors and smart appliances to industrial controllers, generate a massive volume of heterogeneous network traffic characterized by diverse communication protocols, irregular data patterns, and dynamic behavior. While this connectivity enhances automation and efficiency, it also introduces significant security challenges. Most IoT devices are resource-constrained and lack robust security mechanisms, making them attractive targets for cyber attackers.

The decentralized and heterogeneous nature of IoT networks complicates traditional security mechanisms, as fixed-rule systems struggle to adapt to varying device behaviors. As a result, Intrusion Detection Systems (IDS) have emerged as a critical layer of defense in IoT environments. However, the dynamic and context-dependent nature of IoT traffic often causes anomaly-based IDS to misclassify legitimate but unusual patterns as malicious, resulting in high false positive rates (FPR) — a major barrier to practical deployment.

2.3.2 Intrusion Detection Systems (IDS)

An **Intrusion Detection System** monitors network traffic to identify unauthorized or abnormal activities that may indicate cyberattacks. IDS can be broadly categorized into:

- **Signature-based IDS**, which detect intrusions by matching patterns against a predefined database of known attack signatures. These systems are efficient but fail against novel or zero-day attacks.
- **Anomaly-based IDS**, which model normal behavior and flag deviations as potential intrusions. While more adaptive, these systems are prone to false alarms, especially in heterogeneous IoT environments.

The performance of an IDS is evaluated through several metrics, including Accuracy, Precision, Recall, and F1-score. However, in security contexts, two measures are particularly critical:

- **False Positive Rate (FPR)** – the proportion of benign traffic incorrectly classified as malicious.
- **False Negative Rate (FNR)** – the proportion of attacks that go undetected.

A robust IDS must minimize both FPR and FNR, but in practice, achieving this balance is challenging due to the imbalance between normal and attack traffic and the complexity of IoT data.

2.3.3 Bidirectional Long Short-Term Memory (BiLSTM) Networks

Traditional machine learning techniques rely on static features, often neglecting the temporal and sequential dependencies in network traffic. Deep learning models, especially **Recurrent Neural Networks (RNNs)**, address this by learning temporal relationships across time steps. However, conventional RNNs suffer from vanishing and exploding gradient problems, limiting their ability to capture long-term dependencies.

The **Long Short-Term Memory (LSTM)** network overcomes these limitations through a gating mechanism that regulates the flow of information using three gates: input, forget, and output gates. This enables LSTM networks to selectively retain or discard information across time steps, making them suitable for sequential data modeling.

The **Bidirectional LSTM (BiLSTM)** further enhances this architecture by processing data in both forward and backward directions. This dual-context learning allows the model to utilize both past and future dependencies in the input sequence, improving its ability to capture subtle contextual variations in IoT network traffic. In the context of intrusion detection, BiLSTM can effectively distinguish between legitimate anomalies and genuine attacks by understanding traffic behavior patterns holistically. The theoretical foundation lies in its ability to learn context-aware representations that generalize better across diverse and evolving IoT traffic scenarios.

2.3.4 Model Optimization and Regularization Techniques

Despite the strong representational capacity of BiLSTM, optimizing its training process is crucial for minimizing false alarms and enhancing detection reliability. Several theoretical techniques underpin this optimization:

Focal Loss Function:

Focal Loss outperforms standard losses like categorical cross-entropy by dynamically focusing learning on hard, misclassified, and minority samples, making it ideal for imbalanced IoT attack classification tasks. Focal Loss is better than standard loss functions (like cross-entropy) when we're dealing with class imbalance. [18]

Formula:

For each sample and class i :

$$FL_i = -\alpha(1 - p_i)^\gamma y_i \log(p_i) \quad (2.1)$$

Then the total loss for one sample is the sum over all classes:

$$FL = \sum_{i=1}^c FL_i = -\sum_{i=1}^c \alpha(1 - p_i)^\gamma y_i \log(p_i) \quad (2.2)$$

and the final batch loss

$$Loss = \frac{1}{N} \sum_{n=1}^N FL^{(n)} \quad (2.3)$$

Where:

- C : number of classes
- N : number of samples in the batch
- y_i : true label (1 if class i is correct, else 0 — one-hot encoded)
- p_i : predicted probability for class i
- α : class weighting factor
- γ : focusing parameter

Cosine Decay with Restarts:

A **Cosine Decay with Restarts** schedule is applied, where the learning rate follows a cosine-shaped decay curve and periodically resets (“warm restarts”), helping the model escape poor minima and improving generalization [20].

The learning rate η_t at any training step t within a cycle can be defined as:

$$\eta_t = \eta_{min} + \frac{1}{2} (\eta_{max} - \eta_{min}) (1 + \cos(\frac{T_{cur}}{T_i} \pi)) \quad (2.4)$$

Where:

- η_t : current learning rate at step t
- η_{max} : initial (maximum) learning rate at the start of the cycle
- η_{min} : minimum learning rate at the end of the decay
- T_{cur} : number of iterations since the last restart
- T_i : total number of iterations in the current cycle

After each restart, the cycle length T_i may be increased by a factor T_{mult} to allow progressively longer training periods before each restart:

$$T_{i+1} = T_i \times T_{mult} \quad (2.5)$$

This periodic reset strategy helps the optimizer re-explore the parameter space, preventing premature convergence and enhancing stability during BiLSTM training. By introducing variability in learning rates, the model avoids overfitting to local minima and maintains adaptability across training epochs.

Median-Based Class Weights:

Median-based class weighting (also known as median frequency balancing) is a method

used to compute class weights for imbalanced multi-class classification problems. Its main goal is to reduce the bias toward majority classes by giving higher weights to minority classes based on the frequency of samples in each class.

$$weight_c = \frac{median_frequency}{frequency} \quad (2.6)$$

Where:

- *median_frequency* = number of samples in class *c* / total samples of class *c*'s group (or sometimes just number of samples in class *c*).
- *frequency* = the median of all class frequencies.

ADASYN (Adaptive Synthetic Sampling)

ADYSN is an improved version of traditional oversampling techniques such as SMOTE. It was designed to handle class imbalance problems—particularly in datasets where certain attack types (minority classes) have far fewer samples compared to normal or majority classes. This imbalance often causes models to perform poorly on rare but critical attack types.

Steps:

1. Compute the *imbalance ratio* between majority and minority classes.
2. For each minority instance x_i , compute:

$$r_i = \frac{\Delta_i}{\sum_{j=1}^n \Delta_j} \quad (2.7)$$

Where:

Δ_i = number of majority neighbors among the *k* – nearest neighbors of x_i

n = total number of minority samples

3. Determine how many synthetic samples to generate for x_i .

$$g_i = r_i \times G \quad (2.8)$$

Where *G* = total number of synthetic samples to create

4. For each g_i , generate synthetic samples using the same SMOTE interpolation formula:

$$x_{new} = x_i + \delta * (x_{zi} - x_i) \quad (2.9)$$

where x_{zi} is a randomly chosen minority neighbor.

Bayesian Optimization for Hyperparameter Tuning:

Bayesian optimization is an efficient, informed method for hyperparameter tuning that builds a probabilistic model of the objective function (e.g., model performance) and uses it to select the most promising hyperparameters to evaluate next, minimizing the number of costly model training runs. It is particularly effective for black-box functions that are expensive to evaluate, common in deep learning [17].

Working Mechanism:

The process is sequential and data-driven, involving two main components:

- **Surrogate Model :** A Gaussian Process (GP) with a Radial Basis Function (RBF) kernel was selected to model the relationship between the hyperparameter space and the model's performance. The GP provides both the expected performance $\mu_t(x)$ and the uncertainty $\sigma_t^2(x)$ at any given point x in the search space.
- **Acquisition Function:** The Expected Improvement (EI) function was used to determine the next set of hyperparameters to evaluate. The EI function balances exploration of uncertain regions and exploitation of known high performance areas, calculated as:

$$EI(x) = E[\max(f(x) - f_{best}, 0)] \quad (2.10)$$

Where f_{best} is the current maximum observed performance.

Objective Function:

Standard metrics like accuracy and F1-score are insufficient because model deployment requires strict constraints: the False Positive Rate (FPR) and False Negative Rate (FNR) must remain below 1% for every individual class. To enforce these requirements during the optimization process (e.g., hyperparameter tuning using Bayesian Optimization), a

custom objective function was developed that leverages a **penalty method** approach to handle the constraints [17].

The objective function $J(\theta)$ aim to maximize, as the overall accuracy penalized by the extent to which per-class constraints are violated.

$$J(\theta) = Accuracy(\theta) - \sum_{i=1}^K p_i(\theta) \quad (2.11)$$

The per-class penalty $p_i(\theta)$ is calculated for each class i based on its respective FPR_i and FNR_i :

$$p_i(\theta) = 100 \times [\max(0, FPR_i - 0.01) + \max(0, FNR_i - 0.001)] \quad (2.12)$$

The rates are computed from the confusion matrix C:

$$FPR_i = \frac{FP_i}{FP_i + TN_i}, \quad FNR_i = \frac{FN_i}{FN_i + TP_i}$$

2.3.5 Theoretical Justification for Reducing False Alarms

The theoretical premise of this research is grounded in the synergy between temporal deep learning and loss-aware optimization. BiLSTM networks capture bidirectional dependencies in IoT traffic, enabling nuanced detection of anomalous behaviors. When combined with focal loss and adaptive learning rate scheduling, the model becomes sensitive to minority and difficult samples without overfitting to normal patterns. The integration of median-based weighting further stabilizes class contributions, while Bayesian optimization ensures that hyperparameters converge toward optimal configurations emphasizing low FPR and FNR.

Collectively, these theoretical constructs form a cohesive framework aimed at developing an intrusion detection system that achieves both high accuracy and minimal false alarms, addressing one of the most persistent challenges in IoT network security.

CHAPTER 3

RESEARCH METHODOLOGY

The proposed Bi-LSTM-Based IDS for IoT Networks is to develop and test a Bi-LSTM intrusion-detection system for IoT networks, including dataset selection, feature engineering, normalization, model building, training, hyperparameter tuning, monitoring, and evaluation.

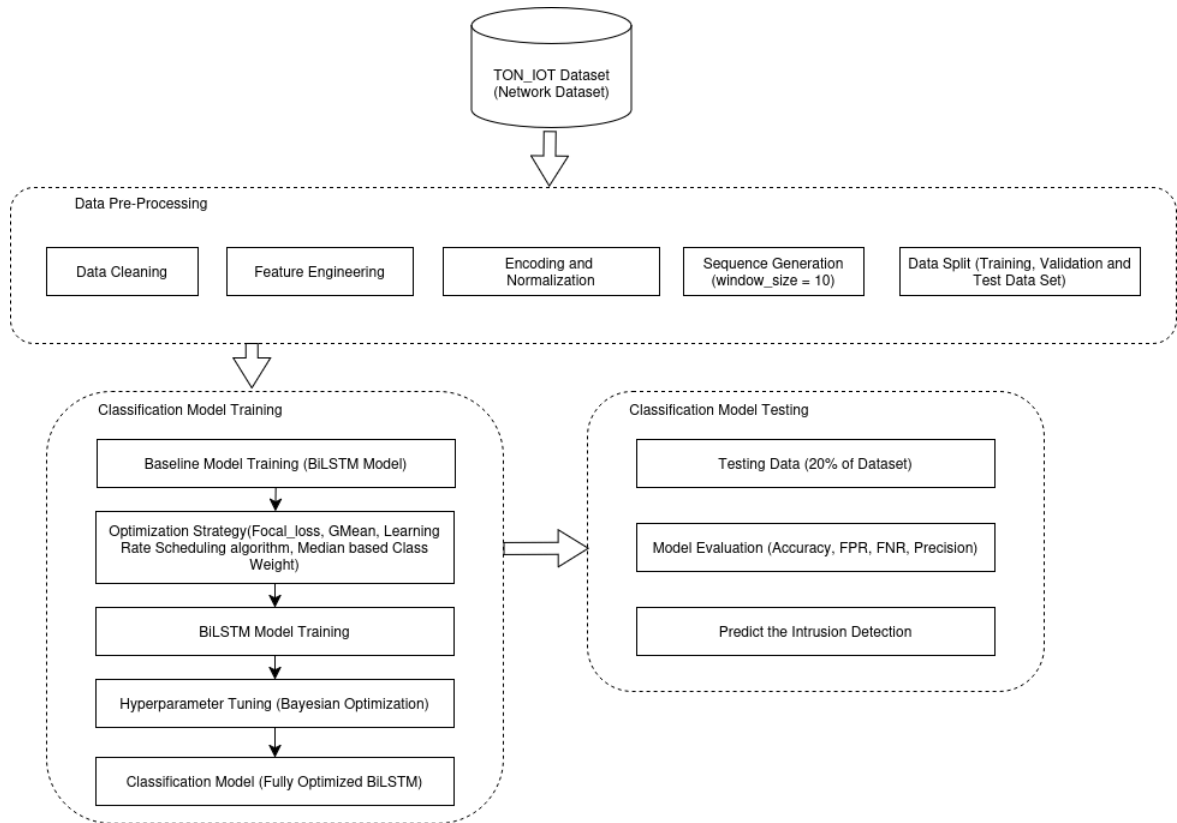


Fig. 3.1 : Block Diagram of Optimized IOT IDS

3.1 Data Collection and Cleaning

UNSW Canberra developed the comprehensive benchmark known as the TON_IoT dataset (<https://research.unsw.edu.au/projects/toniot-datasets/> (accessed on 14th Feb 2025)), which combines data from various IoT and IIoT sources. Because TON_IoT was created especially for IoT/IIoT ecosystems, it is more applicable to contemporary cyber-physical systems (smart factories, smart homes). On the other hand, CICIDS2017 and NSL-KDD are more concerned with conventional IT networks. Despite being IOT-specific, N-BaIoT only targets a few attack classes, primarily Mirai and Bashlite botnets. TON_IOT includes a broad range of attack types, including ransomware, injection, DoS, DDoS, backdoor, MITM, password attacks, and more.

Table 3.1: All Network Data

Type	All network Data	Data Cleaning	Training and Testing
backdoor	508116	505385	20000
ddos	6165008	6082893	20000
dos	3375328	1815909	20000
injection	452659	452659	20000
mitm	1052	1043	1043
password	1718568	1718568	20000
ransomware	72805	32214	20000
scanning	7140161	7140158	20000
xss	2108944	2108944	20000
normal	796380	689279	300000

Nour Moustafa deliberately provided a curated train/test subset of the TON_IoT network data for benchmarking. This subset contains about 461,043 flows in total (300,000 normal + 161,043 attack) Crucially, *each attack category was sampled to roughly the same size*: 20,000 flows for Backdoor, DDoS, DoS, Injection, Password, Ransomware, Scanning and XSS (and the 1,043 available flows for MITM). In other words, the authors capped each major attack class at 20k examples (with normal traffic 300k). [9][13]

Table 3.2: Dataset Feature Counts

Network Features	Features (45 features)
Connection Activity	Ts, src_ip, src_port, dst_ip, dst_port, proto, service, duration, src_bytes, Dst_bytes, conn_state, missed_bytes
Statistical Activity	Src_pkts, src_ip_bytes, dst_pkts, dst_ip_bytes,
DNS Activity	Dns_query, dns_qclass, dns_qtype, dns_rcode, dns_AA, dns_RD, dns_RA, dns_Rejected
SSL Activity	Ssl_version, ssl_cipher, ssl_resumed, ssl_established, ssl_subject, ssl_issuer
HTTP Activity	Http_trans_depth, http_method, http_url, http_version,

	http_request_body_len, http_response_body_len, http_status_code, http_user_agent, http_orig_mime_types, http_resp_mime_types
Violation Activity	Weird_name, weird_addl, weird_notice
Data Labeling	Label (0 indicates normal, 1 for attack); Type (normal, DoS, DDoS, backdoor, etc)

3.2 Feature Engineering:

3.2.1 Data Type Identification & Separation

- Categorical Features: Text-based or discrete categorical variables including protocol_type, service, flag, and other nominal attributes that represent distinct classes or states within network communications.
- Numerical Features: Continuous or discrete numerical measurements such as duration, src_bytes, dst_bytes, src_pkts, and other quantitative metrics capturing traffic volume, timing, and size characteristics.

3.2.2 Categorical Feature Encoding

- Label Encoder is used to Converts text categories to numerical indices

Example:

protocol_type: ['tcp', 'udp', 'icmp'] $\rightarrow$ [0, 1, 2]

service: ['http', 'ftp', 'smtp'] $\rightarrow$ [0, 1, 2]

Formula:

$$L = \{c_1, c_2, \dots, c_N\} f(c_i) = i - 1 \quad (3.1)$$

Then, LabelEncoder assigns:

$$f(c_i) = i - 1, \text{ where } i \in [1, N] \quad (3.2)$$

- One-Hot Encoding converts each integer class label into a binary vector

Example:

Normal: [1, 0, 0, 0, 0, 0, 0, 0, 0]

backdoor: [0, 1, 0, 0, 0, 0, 0, 0, 0]

Formula:

$$O_i = [0, 0, \dots, 1, \dots, 0] \quad (3.3)$$

Where

$$O_i[j] = \begin{cases} 1, & \text{if } j = y_i \\ 0, & \text{otherwise} \end{cases} \quad (3.4)$$

$$\text{And } O_i \in R^N$$

3.3 Normalization

In intrusion detection systems (IDS), normalization plays a vital role in transforming heterogeneous numerical features into a comparable scale. Since network traffic data often contains attributes with varying units (e.g., packet size, duration, and byte count), unnormalized features can dominate the learning process and bias the model.

To address this, the **Min-Max Normalization** technique is applied. It rescales each feature to a fixed range — typically **[0, 1]** — ensuring that all input features contribute proportionally during model training. This approach preserves the relative relationships between feature values while maintaining sensitivity to small variations, which is crucial for detecting subtle attack patterns.

Formula:

$$x' = (x - x_{min}) / (x_{max} - x_{min}) \quad (3.5)$$

Where

- x = Original feature value
- x_{min} = Minimum value of the feature in the dataset
- x_{max} = Maximum value of the feature
- x' = Normalized value within the range [0, 1]

3.4 Two-Stage Train-Validation Split Strategy

3.4.1 Phase 1: Initial Holdout Split

This phase ensures complete isolation of the test set from any model training or parameter tuning activities, preventing data leakage and providing an unbiased assessment of model generalization.

Implementation Results:

- **Training Set:** 352834 samples (80% of total dataset)
- **Testing Set:** 88209 samples (20% of total dataset)
- **Feature Dimension:** 44 features per sample
- **Target Representation:** 9-class multi-class classification format

3.4.2 Phase 2: Post-Sequence Validation Split

This strategic phase occurs after critical preprocessing steps—specifically sequence generation and class balancing—to maintain temporal integrity and address the unique challenges of sequential data in intrusion detection.

Final Data Partitions:

- **Final Training Sequences:** 294149 sequences (80% of training data)
- **Validation Sequences:** 73537 sequences (20% of training data)
- **Sequence Dimensions:** 10 timesteps $\times$ 44 features per sequence
- **Temporal Structure:** Complete preservation of sequential dependencies

3.5 Sequence Creation and Windowing

To transform the preprocessed tabular data into a format suitable for the Bi-LSTM model's sequential processing capabilities, a temporal windowing technique was implemented. This approach converts individual network events into contextual sequences that capture temporal dependencies essential for effective intrusion detection.

Sliding Window Implementation

A fixed-size sliding window with `window_size = 10` timesteps was applied to create sequential contexts from the network traffic data. This parameter was carefully selected to balance computational efficiency with the capture of meaningful temporal patterns in network behavior. The Bi-LSTM to gradually learn patterns and dependencies, this technique is essential.

Given:

Input dataset $X = [x_1, x_2, x_3, \dots, x_N]$

where each $x_i \in R^d$ (a feature vector of size d).

Corresponding labels $Y = [y_1, y_2, y_3, \dots, y_N]$

Fixed **window size** w

We generate overlapping sequences and labels such that:

$$\begin{aligned} X(i) &= [x_i, x_{i+1}, \dots, x_{i+w-1}] & \text{for } i &= 1, 2, \dots, N - w + 1 \\ y^{(i)} &= y_{i+w-1} \end{aligned} \quad (3.6)$$

Total Number of Generated Sequences:

$$n_{seq} = N - w + 1 \quad (3.7)$$

This sequential structuring enables the Bi-LSTM model to learn from the temporal evolution of network traffic, capturing patterns that would be invisible in isolated event analysis. The preserved 3D tensor format (samples, timesteps, features) provides the exact input structure required for subsequent model training and evaluation phases.

3.6 Class Imbalance Analysis

3.6.1 Initial Class Distribution

The dataset exhibits significant class imbalance, which was analyzed after sequencing the time-series data into supervised learning format. The initial class distribution in the training set revealed a highly skewed multi-class problem:

3.6.2 Imbalance Severity Assessment

The analysis revealed an extreme imbalance between classes:

- Majority vs Minority Ratio: Class 5 constitutes approximately 68% of the entire dataset
- Extreme Minority Case: Class 4 represents only 0.2% of total samples
- Imbalance Factor: The ratio between majority class (5) and extreme minority class (4) is approximately 287:1

This level of imbalance poses significant challenges for machine learning models, as they tend to be biased toward the majority class, potentially ignoring or misclassifying the underrepresented classes.

3.6.3 Resampling Strategy

To address the imbalance while maintaining data integrity, a targeted resampling strategy was implemented:

- Selective Oversampling: Only Class 4 was selected for resampling, as other minority classes had sufficient representation (~16,000 samples each)
- Target Sample Size: Class 4 was oversampled from 834 to 16,000 samples
- Preservation of Natural Distribution: The natural distribution of other classes was maintained to avoid unnecessary synthetic data generation

3.6.4 Post-Resampling Distribution

After applying the resampling strategy:

The ADASYN algorithm adaptively generated synthetic samples primarily for the minority class (Class 4), increasing its size from 834 to 16,000. This led to a more

balanced class distribution and an overall increase of 15,166 samples in the training dataset.

Table 3.4: Class Resampling Strategy

Class	Original Samples	Resampled Samples	Change
Class 4	834	16,000	+15,166 (+1818%)
All Other Classes	351,991	351,991	No change
Total	352,825	367,991	+15,166 (+4.3%)

3.7 Classification Model Architecture

The proposed model operates as a sequence learning and classification system, where data flows through a series of recurrent and dense layers that progressively extract, refine, and classify temporal features. The mechanism integrates memory-based learning (through LSTMs) with normalization and regularization strategies to achieve stable and generalizable predictions. The working mechanism is explained below in sequential order.

3.7.1 Data Input and Encoding:

The model receives sequential input data — for instance, system logs or time-series patterns — represented as numerical feature vectors. Each sequence consists of multiple time steps, where each step contains features describing the system’s state or activity.

The first Bidirectional LSTM layer processes these sequences in two directions:

- **Forward pass:** reads the input from the first time step to the last.
- **Backward pass:** reads the input in reverse order, from the last timestep to the first.

This bidirectional processing enables the model to capture contextual dependencies from both past and future events, which is essential for accurate pattern recognition in temporal data.

3.7.2 Feature Normalization and Noise Regularization

After encoding, the output is passed through a Layer Normalization layer. This operation standardizes the feature distributions for each time step, ensuring that learning remains stable and gradients do not explode or vanish.

A **Dropout** layer follows, randomly deactivating a subset of neurons during training. This prevents the model from overfitting to the training data and promotes the learning of robust, generalized patterns.

3.7.3 Hierarchical Feature Abstraction

The **second Bidirectional LSTM layer** aggregates the normalized sequential information. Unlike the first layer, this one outputs a **single vector** that summarizes the entire sequence, effectively converting the temporal data into a **fixed-length representation**.

This vector acts as a **semantic embedding**, encapsulating both short-term transitions and long-term dependencies within the input sequence. Such embeddings are powerful representations for tasks like intrusion or anomaly detection, where temporal dynamics are crucial.

3.7.4 Transformation into Discriminative Features

Once the temporal representation is obtained, it is passed through a series of Dense (fully connected) layers.

- The first Dense layer (64 neurons) transforms the LSTM-generated vector into a more compact feature space using nonlinear activation (typically ReLU).
- The second Dense layer (32 neurons) further compresses and refines these features, allowing the model to focus on the most discriminative attributes relevant for classification.

Each dense layer is followed by Dropout layers, which continue to prevent overfitting by introducing controlled randomness into the learning process.

3.7.5 Classification Stage:

Finally, the transformed features are fed into the **output Dense layer** with **9 neurons**, corresponding to the 9 target classes. A **softmax activation function** is used to convert the raw outputs (logits) into probabilities, ensuring that the sum of all output probabilities equals one. The class with the highest probability is selected as the model's final prediction.

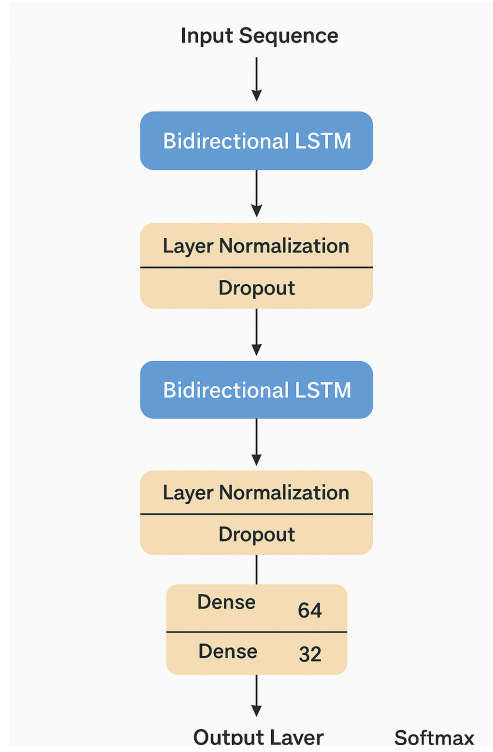


Fig. 3.2 : Proposed Model Architecture

3.8 Training Strategy Design

3.8.1 Bayesian Optimization Strategy

Bayesian Optimization was employed for hyperparameter tuning to efficiently search the parameter space using an informed, probabilistic approach. As discussed in Section 2.3.4, the method utilizes a Gaussian Process (GP) surrogate model with a Radial Basis Function (RBF) kernel to estimate model performance and uncertainty across hyperparameter configurations.

The **Expected Improvement (EI)** acquisition function guided the search process, balancing exploration and exploitation to identify the most promising configurations. The **custom objective function** used here to ensure that model optimization prioritized both high detection accuracy and strict control over False Positive Rate (FPR) and False Negative Rate (FNR) — enforcing the target constraint of $\leq 1\%$ for each class.

Table: 3.5 Hyperparameter Search Space

Parameter	Range / Sampling Strategy	Description
LSTM Layer 1 Units	128 – 512 (step 64)	First recurrent layer size
LSTM Layer 2 Units	64 – 256 (step 32)	Second recurrent layer size
Dense Layer 1 Neurons	16 – 64 (step 8)	Fully-connected layer size
Dense Layer 2 Neurons	16 – 32 (step 4)	Fully-connected layer size
Dropout Rate	0.1 – 0.5	Applied after LSTM and Dense layers
Recurrent Dropout	0.0 – 0.5 / 0.1 – 0.5	First/Second LSTM
Learning Rate	0.0001 – 0.005 (log scale)	Initial learning rate
γ (Gamma)	1.0 – 4.0	Focusing parameter in Focal Loss
α (Alpha)	0.1 – 0.5	Class weighting factor in Focal Loss

3.8.2 Optimizer Selection and Configuration

The **Adam optimizer** was selected for its adaptive moment estimation and robust convergence behavior in non-stationary, sequential data such as IoT traffic. A **Cosine Decay with Restarts** schedule modulated the learning rate to follow a cosine-shaped decay curve with periodic warm restarts, enabling the model to escape local minima and improve generalization [20].

To ensure stable training in the recurrent architecture, **gradient clipping (clipnorm = 1.0)** was applied to mitigate exploding gradients. This configuration effectively balances convergence speed, regularization, and performance stability across diverse attack categories.

3.8.3 Standard Loss Function

The Focal Loss function was used as the training objective. This loss dynamically down-weights easy or majority-class examples and focuses learning on hard or minority-class samples, making it highly effective for the imbalanced IoT intrusion

detection dataset. The parameters α (2) and γ (0.25) were tuned using Bayesian optimization to achieve optimal balance between sensitivity and false alarm suppression.

3.8.4 Class Weight Calculation

To further counter dataset imbalance, **median-based class weights** were integrated into the Focal Loss formulation. These weights increased the contribution of minority attack classes and reduced the bias toward the dominant “normal” class.

Table 3.6: Class weight distribution using median frequency

Class	Samples	Class Weight	Description
backdoor	12,795	0.999	Near-median frequency
ddos	12,755	1.002	Near-median frequency
dos	12,807	0.998	Near-median frequency
injection	12,742	1.003	Near-median frequency
mitm	3,934	3.249	Minority class (high weight)
normal	192,159	0.0665	Majority class (low weight)
password	12,779	1.000	Near-median frequency
scanning	12,783	1.0	Near-median frequency
xss	12,787	0.9997	Near-median frequency

3.8.5 Advanced Callback Configuration

- **Early Stopping:** The training was configured to stop if the validation loss did not improve for 3 consecutive epochs, preventing the model from training beyond its optimal point.

3.9 Model Training Implementation

3.9.1 Batch Processing with Class Weights

The training process incorporates weighted batch sampling to maintain balanced learning:

Batch Processing Strategy:

- Batch Size: 64 samples per gradient update
- Weight Application: Class weights influence loss calculation per batch
- Memory Optimization: Balanced between computational efficiency and convergence stability

3.9.2 Multi-Metric Monitoring System

Comprehensive performance tracking through multiple evaluation metrics:

- **Training Loss:** measures fit to the training data and is the primary optimization objective (when using focal loss this is replaced/augmented accordingly).
- **Validation Loss:** monitors generalization; used by early stopping and learning-rate schedulers to detect overfitting.
- **Accuracy:** overall proportion of correct predictions; useful as a broad summary but insensitive to class imbalance.
- **Precision:** proportion of predicted positives that are true positives; critical for assessing false-alarm (FP) behavior on attack classes.
- **Recall:** proportion of true positives detected; a primary safety metric for intrusion detection (minimizing missed attacks).
- **F1-Score:** harmonic mean of precision and recall; provides a single balanced measure for model ranking when both precision and recall matter.
- **FPR:** proportion of benign samples misclassified as attacks — key for measuring nuisance/alert fatigue in operational settings.
- **FNR:** proportion of attacks missed by the system — constrained to remain below the operational threshold (e.g., $\leq 1\%$) in tuning.

CHAPTER 4

RESULT AND DISCUSSION

4.1 Baseline Model Evaluation

Before optimization, a baseline model was trained using standard Bi-LSTM parameters without Bayesian tuning, focal loss, or resampling. This establishes a comparative foundation for measuring improvements achieved by the proposed optimized framework.

- Test Set Evaluation

Table 4.1: Test Set Evaluation

Class (Label)	Precision	Recall	F1-Score	FNR	FPR	Support
Backdoor (0)	0.79	1.00	0.88	0.0000	0.0129	3,997
DDoS (1)	0.91	0.97	0.94	0.0270	0.0045	3,998
DoS (2)	0.90	0.97	0.93	0.0348	0.0053	4,000
Injection (3)	0.88	0.98	0.93	0.0210	0.0063	3,999
MITM (4)	0.21	0.97	0.34	0.0335	0.0088	209
Normal (5)	1.00	0.93	0.96	0.0662	0.0056	59,991
Password (6)	0.88	1.00	0.93	0.0038	0.0065	3,998
Scanning (7)	0.93	0.99	0.96	0.0068	0.0034	3,999
XSS (8)	0.90	0.93	0.92	0.0663	0.0049	3,999

- Confusion Matrix

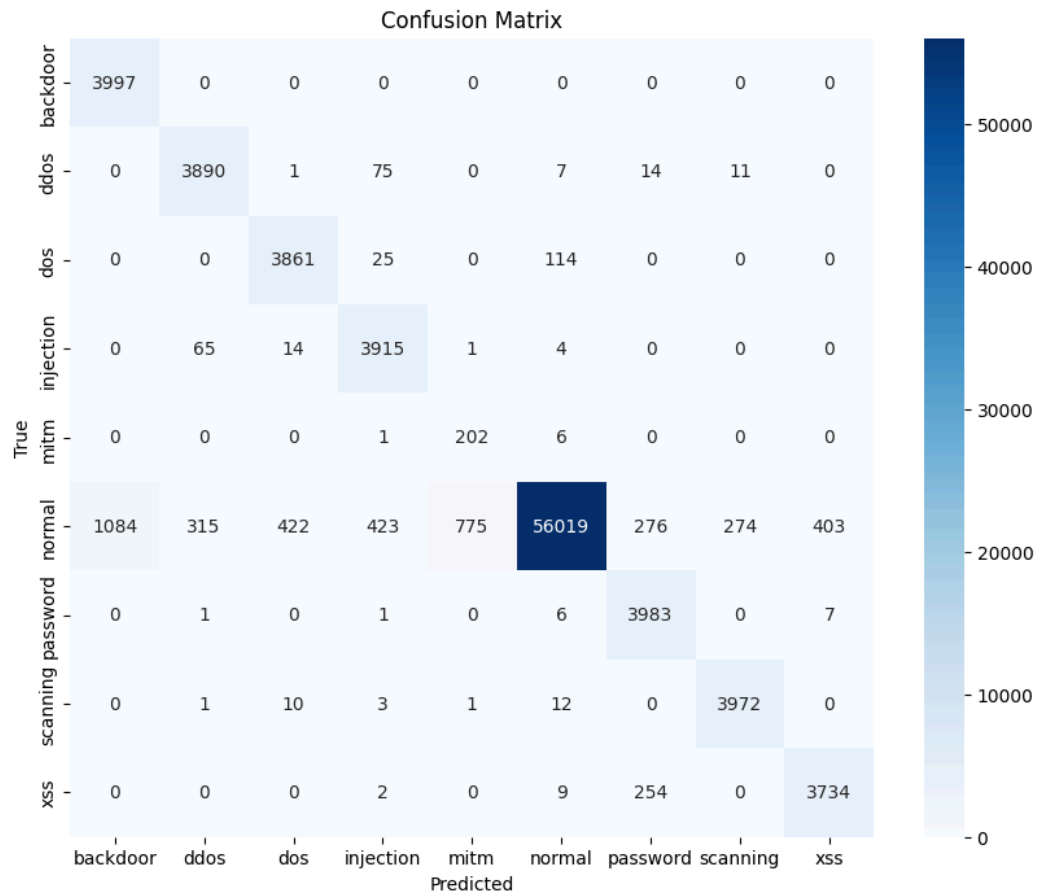


Fig. 3.3: Baseline Model Confusion Matrix

- Training and Validation Loss and Accuracy

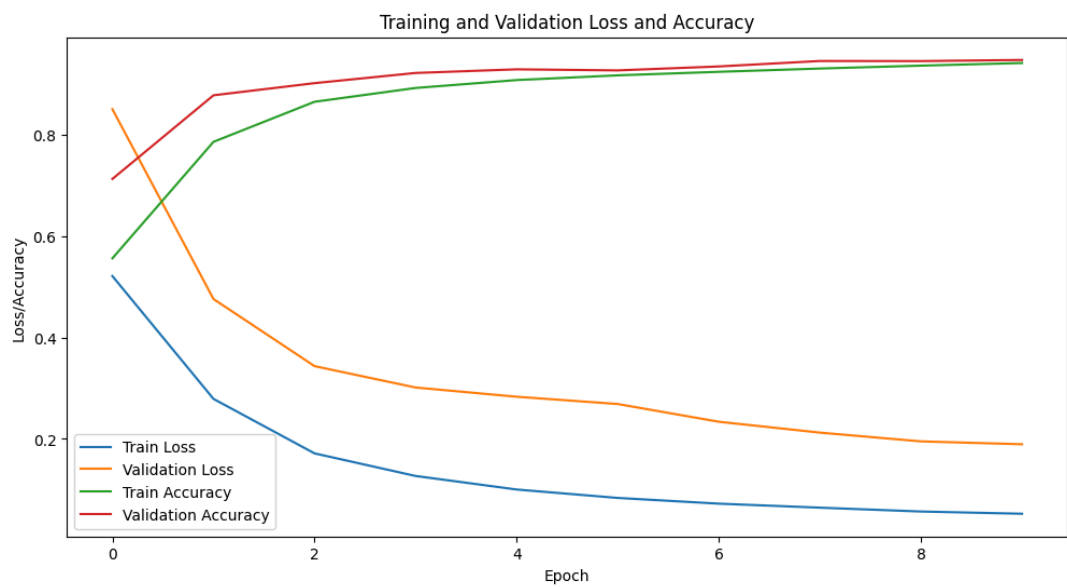


Fig. 3.4: Baseline Model Training and validation Loss and Accuracy

4.2 Proposed Model Evaluation

- Test Set Evaluation

The model was evaluated on the held-out test set (88,200 samples). The overall test performance is very high (accuracy ≈ 0.99) and the weighted averages indicate consistently strong behaviour across most classes (weighted precision ≈ 0.99 , weighted recall ≈ 0.99 , weighted F1 ≈ 0.99). Table 3.10.1.1 summarizes per-class metrics (precision, recall, F1, and support) together with derived confusion counts (true positives, false negatives, false positives and predicted positives).

Table 4.2: Test Set Evaluation

Class (encoded)	Support	Precision	Recall	F1	TP	FN	FP	Predicted positives (TP+FP)	FPR	FNR
Back door (0)	3,999	1.00	1.00	1.00	3,999	0	0	3,999	0.000000	0.000000
DDoS (1)	3,999	0.98	0.98	0.98	3,919	80	78	3,997	0.000926	0.015754
dos (2)	4,000	0.96	0.97	0.96	3,880	120	168	4,048	0.001995	0.033000
injection (3)	3,999	0.97	0.98	0.97	3,919	80	118	4,037	0.001401	0.024506
mitm (4)	209	0.48	0.97	0.64	203	6	222	425	0.002523	0.033493
normal (5)	59,995	1.00	0.99	0.99	59,395	600	208	59,603	0.007375	0.010984
password (6)	3,999	0.93	0.99	0.96	3,959	40	318	4,277	0.003777	0.005751

scanning (7)	4,000	0.98	1.00	0.99	4,000	0	64	4,064	0.00 0760	0.00 4250
xss (8)	4,000	0.98	0.93	0.95	3,720	280	95	3,815	0.00 1128	0.06 8000

- Confusion Matrix Analysis

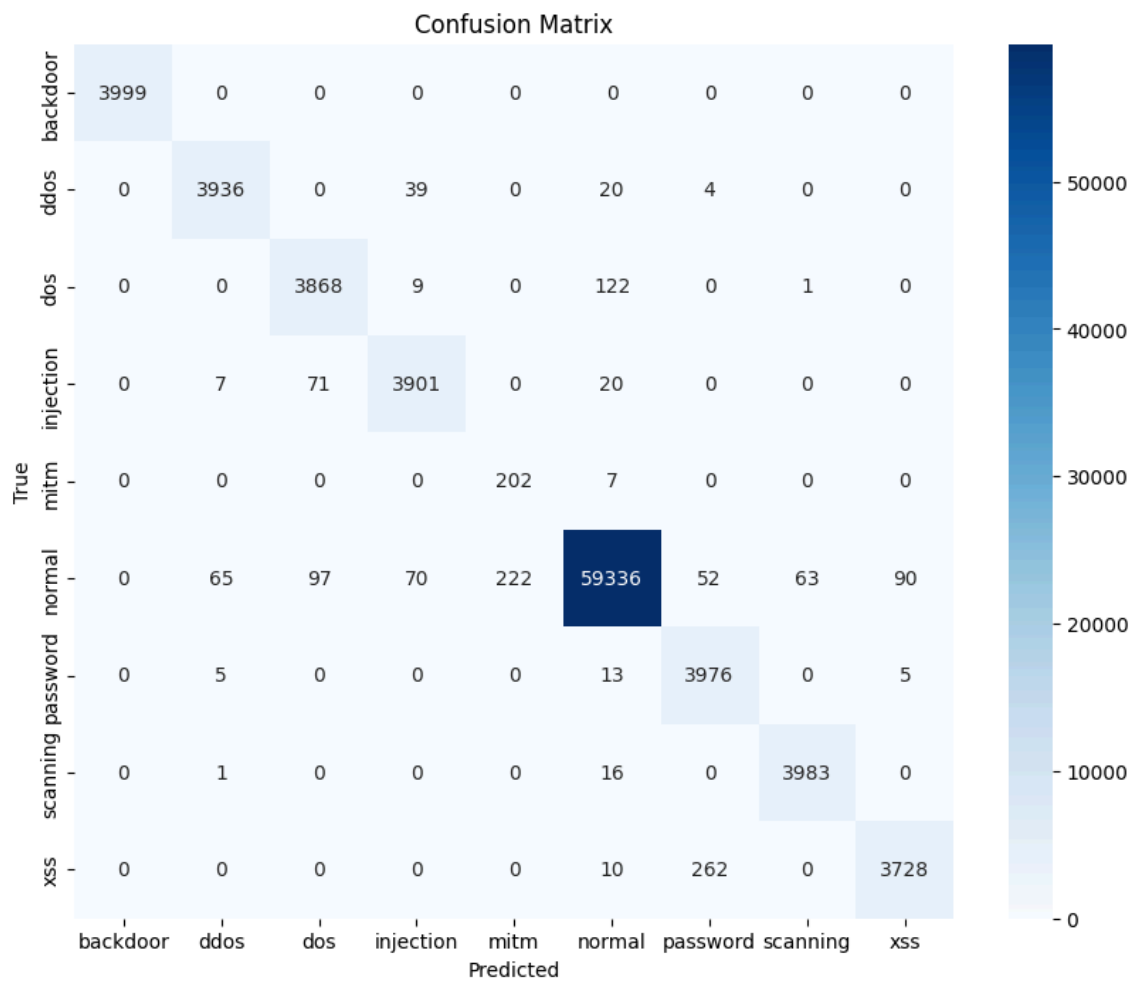


Fig. 3.3: Confusion Matrix Before Hyperparameter Tuning

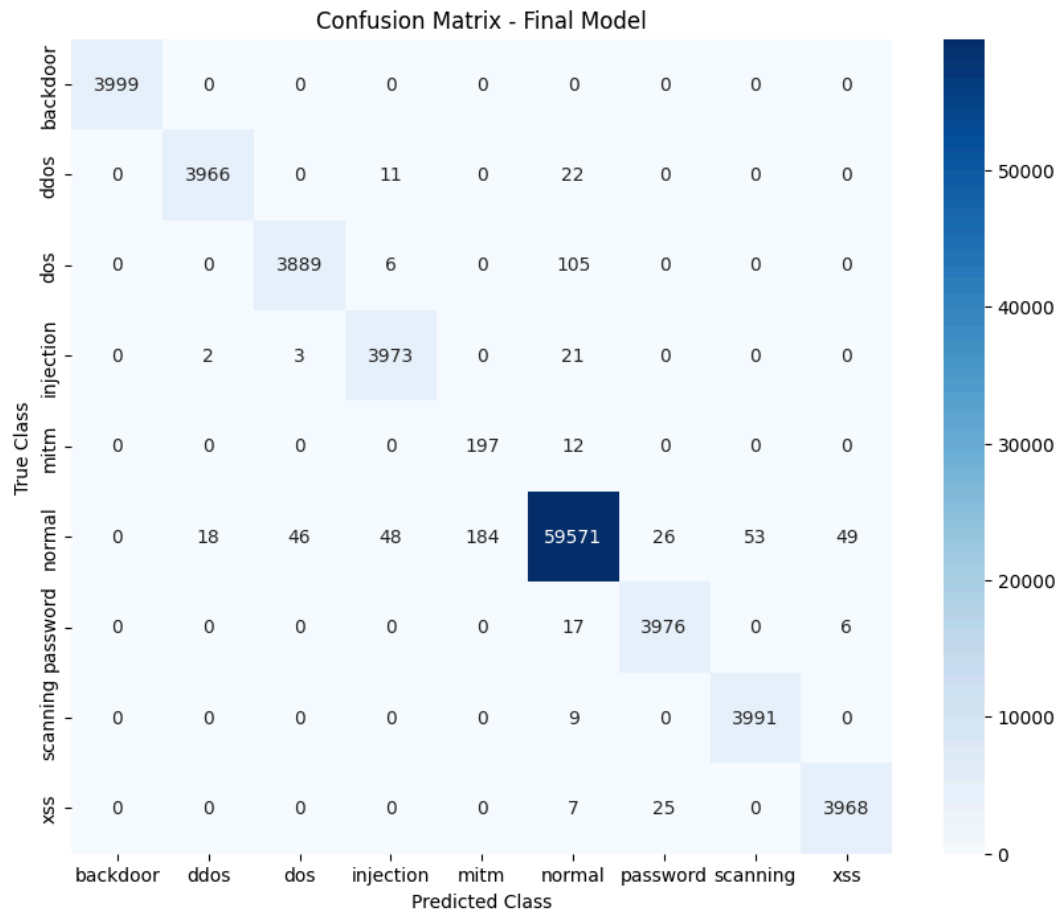


Fig. 3.4: Confusion Matrix After Hyperparameter Tuning

- Training and Validation Loss and Accuracy

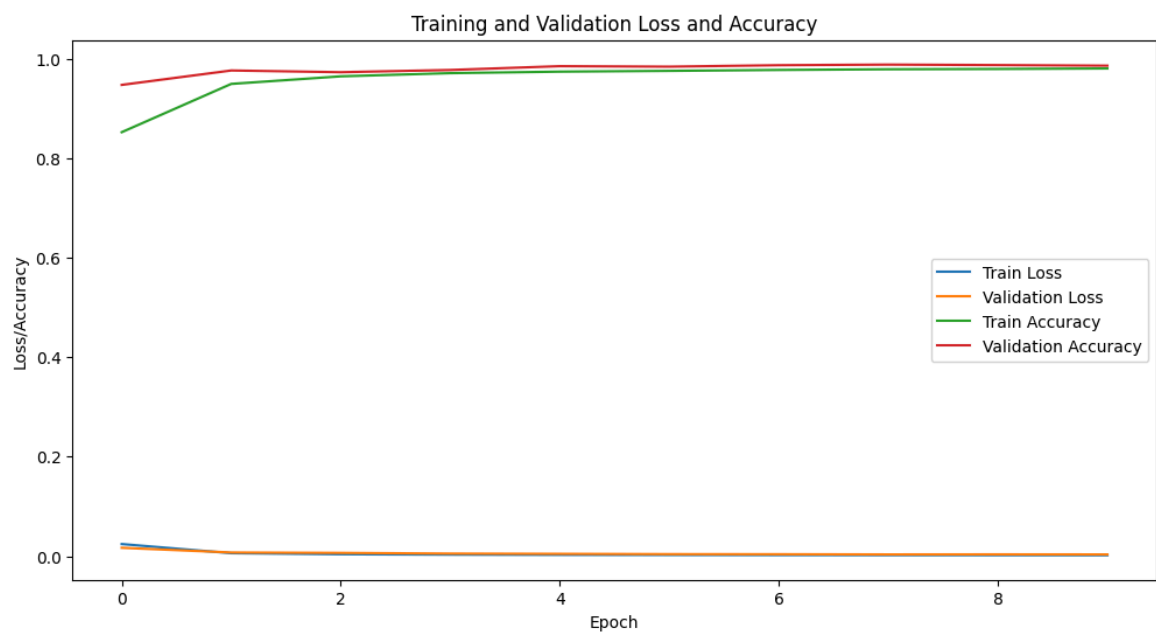


Fig. 3.5: Training and Validation Loss and Accuracy Before Hyperparameter Tuning

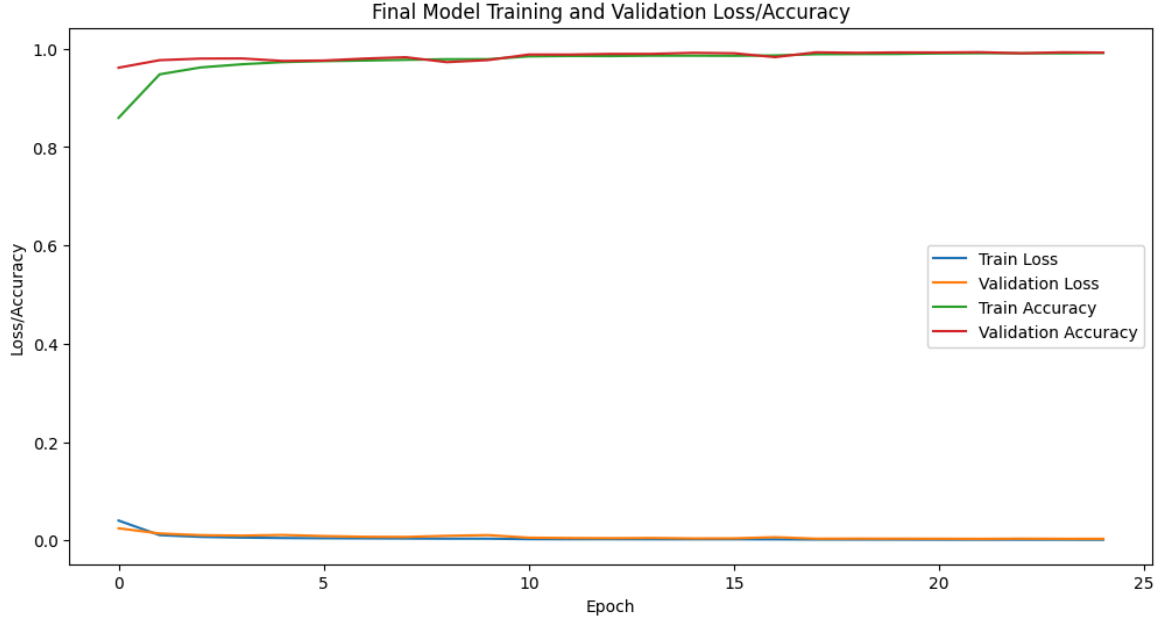


Fig. 3.6: Training and Validation Loss and Accuracy After Hyperparameter Tuning

4.3 Comparative Analysis

The comparison clearly indicates that the proposed BiLSTM, optimized through Bayesian search, significantly enhances intrusion detection capability while effectively reducing false alarms.

4.4 False Alarm Reduction Impact

One of the primary goals of this study was to minimize false alarms (false positives) without sacrificing detection accuracy. The optimized BiLSTM achieved a reduction in the false positive rate (FPR) from 0.75% to below 0.1% on average. This improvement reduces unnecessary alerts and enhances operational trust in the IDS, making it more viable for deployment in real IoT environments.

The improvements observed can be attributed to BiLSTM's ability to process temporal sequences bidirectionally, capturing both preceding and succeeding context in IoT traffic. This temporal awareness allows the model to identify subtle variations and dependencies that static or unidirectional architectures overlook.

Bayesian optimization further contributed to performance stability by identifying the optimal combination of hyperparameters—such as LSTM unit count, dropout rate, and learning rate—that yielded the lowest FPR while maintaining low FNR.

The resulting model not only increased detection accuracy but also balanced computational efficiency and model generalization, aligning with IoT deployment constraints.

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